

Adaptive Multigrid Finite State Projection for the Reaction-Diffusion Master Equation

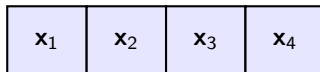
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The Reaction-Diffusion Master Equation

Setting: Domain Ω partitioned into K voxels of size h . State $\mathbf{x}_k$ = molecule counts in voxel k .

diffusion: rate $d = D/h^2$ per molecule



reactions: propensity $\alpha_r(\mathbf{x}_k)$ in each voxel

$$\frac{dp}{dt} = A_{\text{RDME}} p = \underbrace{(A_{\text{rxn}})}_{\text{slow}} + \underbrace{(A_{\text{diff}})}_{\text{fast}} p$$

FSP bottleneck [Munsky & Khammash 2006]

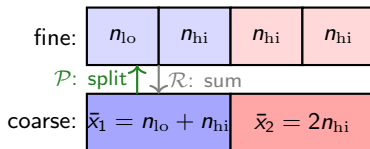
State space $|\mathcal{S}| \sim n_{\text{max}}^K$ grows exponentially in K .

$K = 4:$ $13^4 \approx 28\text{k}$ tractable

$K = 8:$ $13^8 \approx 8 \times 10^8$ intractable

Spatial Coarsening via Lumpability

Idea: merge adjacent fine-voxel pairs into a single coarse voxel. Coarse count:
 $\bar{x}_j = x_{2j-1} + x_{2j}$.



Why it works: intra-pair diffusion at rate D/h^2 equilibrates molecules within a pair *much faster* than inter-pair dynamics. After this fast equilibration the within-pair split follows:

$$\pi(m \mid \bar{x}) = \text{Bin}(\bar{x}, \frac{1}{2})$$

Coarse generator $\bar{A} = \mathcal{R}A_{\text{RDME}}\mathcal{P}$ is itself an RDME on a $K/2$ -voxel grid.

Two-Level V-Cycle

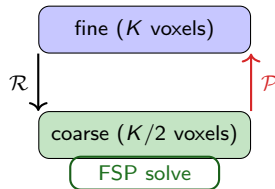
$$p(t) \rightarrow p(t + \Delta t)$$

- 1 **Restrict:** $p^c \leftarrow \mathcal{R} p$
marginalize over within-pair splits
- 2 **Coarse solve:** $\bar{p}^c \leftarrow e^{\Delta t \bar{A}} p^c$
full time step on the small $K/2$ problem
- 3 **Prolong:** $p_{\text{new}} \leftarrow \mathcal{P} \bar{p}^c$
reconstruct fine distribution via $\text{Bin}(\bar{x}, \frac{1}{2})$

Why it's fast:

Coarse state space is $\sim n_{\text{max}}^{K/2}$ vs. n_{max}^K .

$K=8 \rightarrow 4$: 13^4 vs. 13^8 (30k $\times$ smaller).



Analogy to geometric MG:

Restrict = marginalize

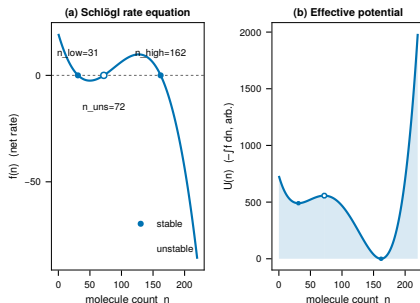
Coarse solve = lumped CME

Prolong = Binomial split

Smooth = intra-pair diffusion

When Coarsening Goes Wrong: The Symmetry Trap

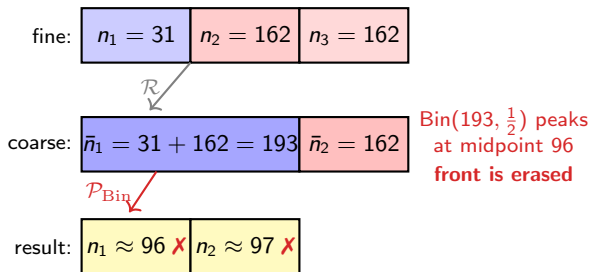
Schlögl model is bistable: two stable states at $n_{\text{low}} = 31$ and $n_{\text{high}} = 162$.



Left: rate equation $f(n)$; zeros are fixed points. **Right:** potential $U(n)$; two wells separated by a barrier at $n = 72$.

The Symmetry Trap

Problem: a spatial front lives *inside* a coarse voxel.



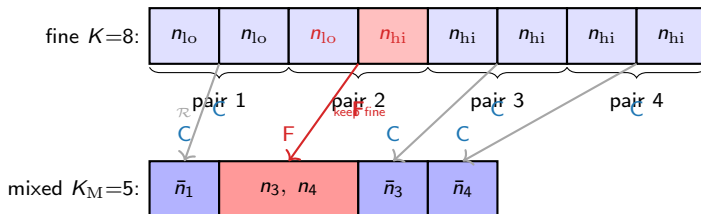
Root cause

The coarse variable $\bar{n}_1 = 193$ is consistent with both $(31, 162)$ and $(162, 31)$. Binomial splits them 50/50 around the midpoint. **Directionality is lost.**

The Fix: Don't Coarsen the Front

Key insight

The Binomial approximation is only wrong *where the distribution is asymmetric within a pair*. For bulk pairs far from the front it works well. **Solution:** selectively refine only the pairs that need it.



Pair 2 (front pair) is kept **F**ine. Bulk pairs 1, 3, 4 are **C**oarsened. Mixed dimension $K_M = 8 - 3 = 5$.

Detecting Which Pairs Need Refinement

Two admissibility diagnostics, computed from voxel means $\mu_k = \mathbb{E}[n_k]$:

Within-pair asymmetry α_j :

$$\alpha_j = \frac{|\mu_{2j-1} - \mu_{2j}|}{\mu_{2j-1} + \mu_{2j}}$$

Large when *the front lies inside pair j* (one voxel high, other low).

Inter-pair boundary gradient β_j :

$$\beta_j = \frac{|\mu_{2j} - \mu_{2j+1}|}{\mu_{2j} + \mu_{2j+1}} \quad (j < K/2)$$

Large when *the front lies on the boundary between pairs j and j + 1*.
Forces both pairs j and j + 1 fine.

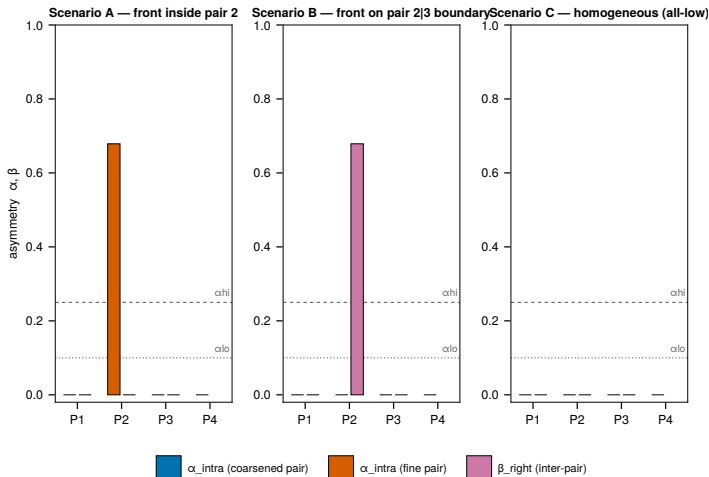
Refinement rule (with hysteresis)

Define $\alpha_j^{\text{eff}} = \max(\alpha_j, \beta_{j-1}, \beta_j)$ (boundary terms $\beta_0 = \beta_{K/2} = 0$).

$$\alpha_j^{\text{eff}} > \alpha_{\text{hi}} \Rightarrow \text{force fine}; \quad \alpha_j^{\text{eff}} < \alpha_{\text{lo}} \Rightarrow \text{coarsen}.$$

Between the bands: keep previous mask (prevents chatter at moving fronts).

Three Scenarios, One Criterion



A: front inside pair 2 (α fires). **B:** front on pair 2|3 boundary (β fires on both). **C:** homogeneous — all coarsened.

The Mixed-Grid CME

Once the mask is chosen, we need a generator for the mixed-dimension state space $\mathcal{S}^M \subset \mathbb{Z}_{\geq 0}^{K_M}$.

Mixed-grid rates:

- **Fine pair** ($\text{mask}_j = 0$): standard per-voxel Schlögl rates; intra-pair diffusion included.
- **Coarse pair** ($\text{mask}_j = 1$): reaction rates with Binomial first-moment closure (e.g. autocatalysis $\propto \bar{n}(\bar{n} - 1)/4$ for a merged pair of equal sub-voxels).
- **Cross-pair diffusion**: fine side $\rightarrow$ coarse side at rate $d n_{2j}$; coarse side $\rightarrow$ fine side at $d \bar{n}_j/2$ (first-moment: $\mathbb{E}[n_{2j+1} \mid \bar{n}_j] = \bar{n}_j/2$).

Why no multiplicative correction is needed

Front pairs are kept fine: their within-pair distribution is never approximated by Binomial. The coarse pairs are bulk pairs where the Binomial assumption holds by symmetry. **Binomial prolongation is then accurate where applied.**

$K = 8$ Selector Recovery at Representative States

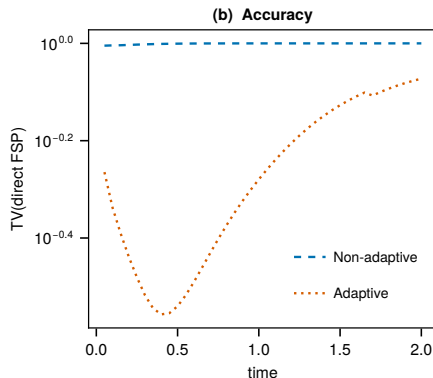
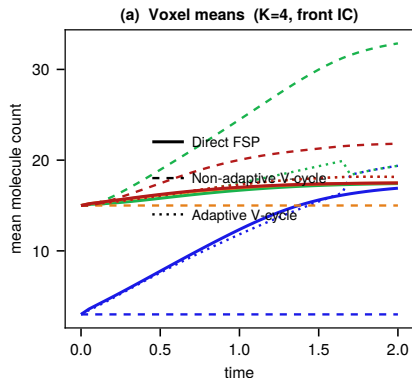
Claim: the criterion fires correctly regardless of where the front falls relative to the coarsening partition.

Scenario	$\alpha_{\text{intra}}[2]$	$\beta_{\text{right}}[2]$	Fine pairs	OK?
A: front inside pair 2	0.68	0.00	{2}	✓
B: front on pair 2 3 bdry	0.00	0.68	{2,3}	✓
C: homogeneous (all low)	0.00	0.00	(none)	✓

$K = 8$ Schlögl ($n_{\text{low}} = 31$, $n_{\text{high}} = 162$). Mask computed from actual depth-1 FSP expansions around each IC. Thresholds $\alpha_{\text{lo}} = 0.10$, $\alpha_{\text{hi}} = 0.25$ throughout.

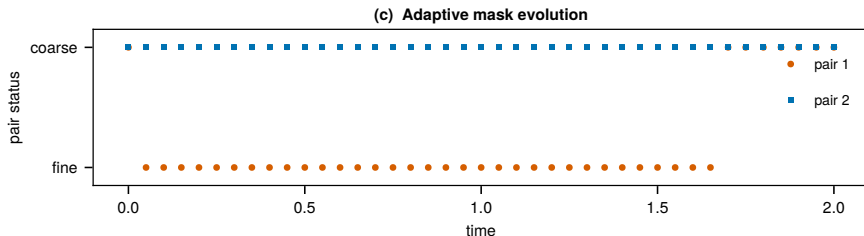
We validate on representative expanded states rather than full trajectories: time-resolved $K = 8$ FSP is itself intractable — that is the problem being solved. The β criterion is what handles case B; without it, pair 3 would be coarsened while straddling the front.

Adaptive V-Cycle: Accuracy on $K = 4$ Schlögl Front



IC: $(n_{\text{low}}, n_{\text{high}}, n_{\text{high}}, n_{\text{high}})$, front inside pair 1. (a) Voxel means: direct FSP (solid), non-adaptive (dashed), adaptive (dotted). (b) TV distance to direct FSP.

$K = 4$ Time-Resolved Mask Evolution



Pair 1 (orange circles) is **forced fine** throughout: the front stays inside it.
Pair 2 (blue squares) is **coarsened** immediately: the bulk is symmetric.

The mask is set automatically at each step from the current distribution. No look-ahead, no hand-tuning of IC or grid alignment.

What we built

- **Probabilistic two-level multigrid** for the CME. Restriction = exact marginalisation. Prolongation = Binomial conditional. Coarse operator = Galerkin-projected RDME.
- **Adaptive pair selection.** Two diagnostics: within-pair α_j and inter-pair β_j . Hysteresis prevents chatter.
- **Mixed-grid CME.** Fine and coarse pairs in a single generator; first-moment closure at cross-pair boundaries.

Key results

- Adaptive mask is **partition-independent**: detects fronts inside a pair (α) *and* on pair boundaries (β).
- Front pairs kept fine $\Rightarrow$ **no symmetry trap**, no special correction needed.
- $K = 8$ Schlögl: direct FSP intractable; adaptive V-cycle solves with modest state growth.

Open directions

- Multi-level hierarchy ($K \rightarrow K/2 \rightarrow \dots$)
- Multi-species / 2D spatial domains
- Error-driven automatic threshold selection

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- Munsky & Khammash (2006). The finite state projection algorithm for the solution of the chemical master equation. *J. Chem. Phys.*, 124:044104.
- Kemeny & Snell (1960). *Finite Markov Chains*. Van Nostrand.
- Briggs, Henson, McCormick (2000). *A Multigrid Tutorial*, 2nd ed. SIAM.
- Dendukuri, Chandrasekaran, Petzold (2025). Adaptive multigrid FSP for the RDME. *In preparation*.